

RAMCO INSTITUTE OF TECHNOLOGY
Department of Mechanical Engineering
Academic Year 2019 – 2020 (Odd Semester)

Innovative practice(s) description

Course code and title: ME8391 Engineering Thermodynamics

Name of the Instructor(s): Mr.M.Sivagaminathan alias Balaji, AP (Sr.Gr.)/Mechanical
Mr.R.Arun Kumar, AP//Mechanical

Date & Time: 19.10.2019 & 9:30AM to 11:30AM

Collaborative Learning on Rankine, Reheat and Regenerative Cycle problems

Topic: Problem solving on Rankine, Reheat and Regenerative Cycles

Justification for the topic:

There are three reasons why we chose this topic for collaborative learning:

- During end semester examination, 30 marks out of 100 will be asked from this topic (one each in 2 marks, 13 marks and 15 marks)
- Rankine cycle taught in the subject Engineering Thermodynamics (III Semester) will be implemented in the subjects Thermal Engineering II (V Semester) and Power Plant Engineering (VII Semester)
- Student's attainment level in the topic in their Internal Assessment Test – II (IAT – II) was not up to the expected levels

I thought that implementation of collaborative learning will be helpful to make the students learn the topic in an effective way with better outcome than the conventional teaching

Team Style: 1 Fast learner (student who scored 80+ marks in IAT – II)

2 Average learners (students who scored between 50 and 80 marks in IAT – II)

1 Slow learner (student who scored less than 50 in IAT – II)

Students were grouped into different teams with 4 in each group. They were asked to prepare the topic and again an input was given to the students about problem solving procedure. Then each team was asked to solve three different problems. Once all the teams solved the problems they shared their problems with other teams.

Outcome:

Through this the fast learner in the group helped his slow learning peer in solving the problem. All the members had solved all the given problems to their team.

This activity helps in attaining the following CO – PO, PSO mapping:

Course Outcome / Programme Outcome	PO1	PO2	PO4	PO9	PO10	PSO3
CO3 – Student will be able to apply Rankine cycle to steam power plant and compare few cycle improvement methods	3	3	1	1	1	3

